



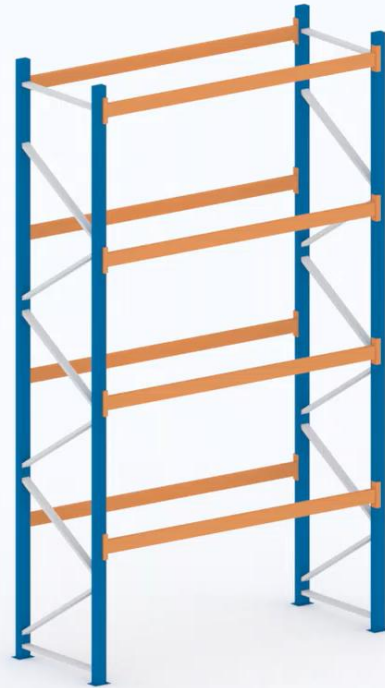
**HACKUPC 2026:
WAREHOUSE OPTIMIZER**

ANATOMY OF A RACK



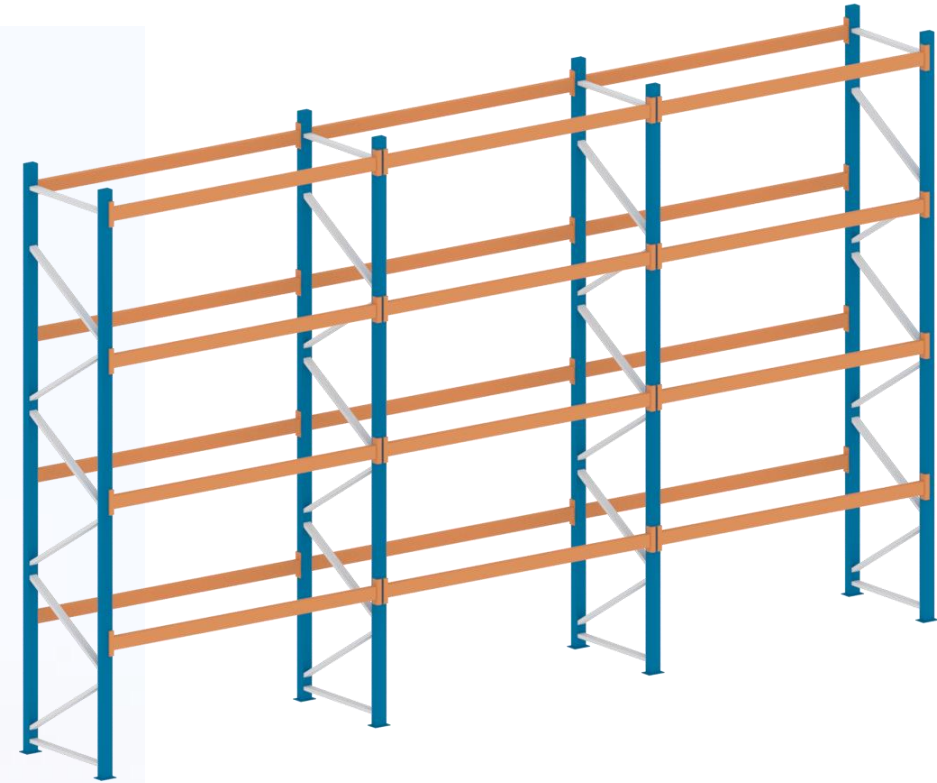
Frames + Beam =

Bay:

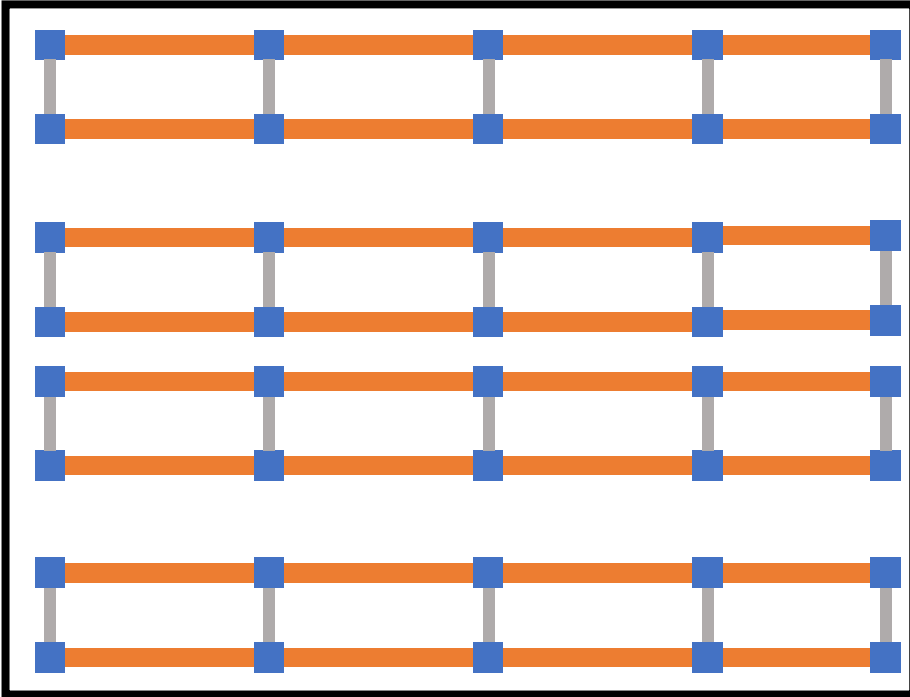


Many Bays =

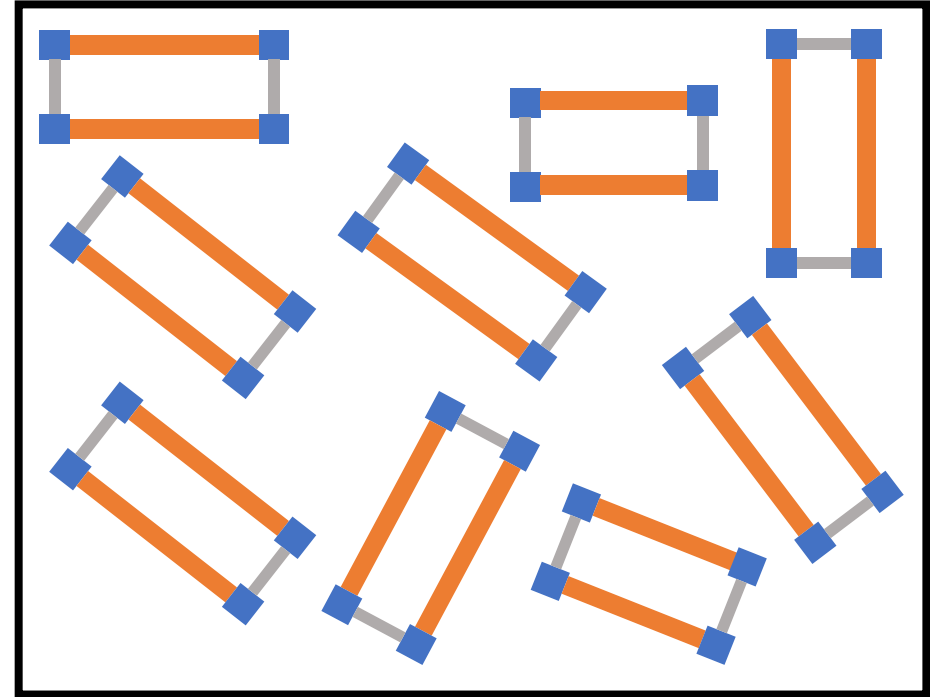
Row:



OUR PROBLEM



or

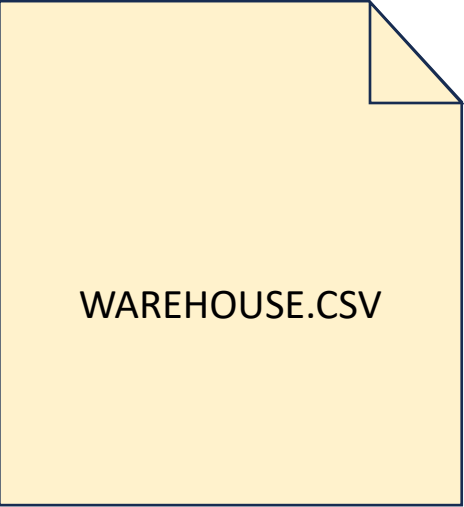


?

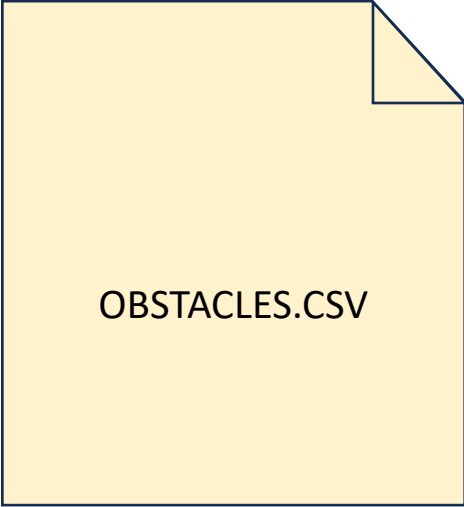
Your goal: **Place bays** in a warehouse the **cheapest way** using the largest ammount of area

INPUT

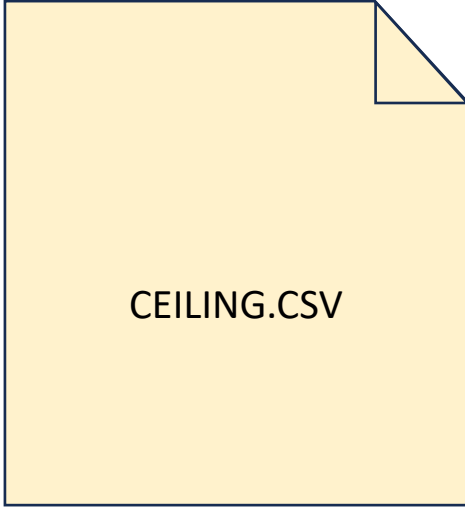
Each case consists on 4 CSV files



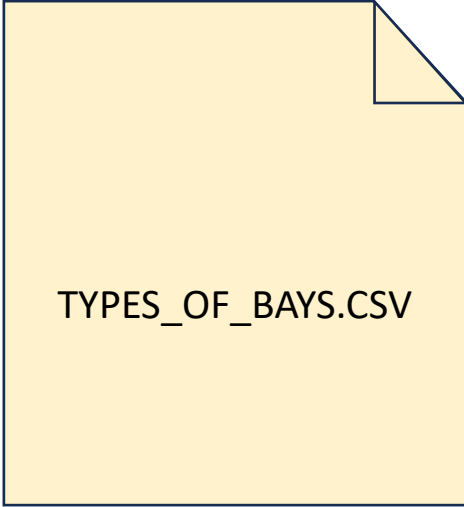
WAREHOUSE.CSV



OBSTACLES.CSV



CEILING.CSV

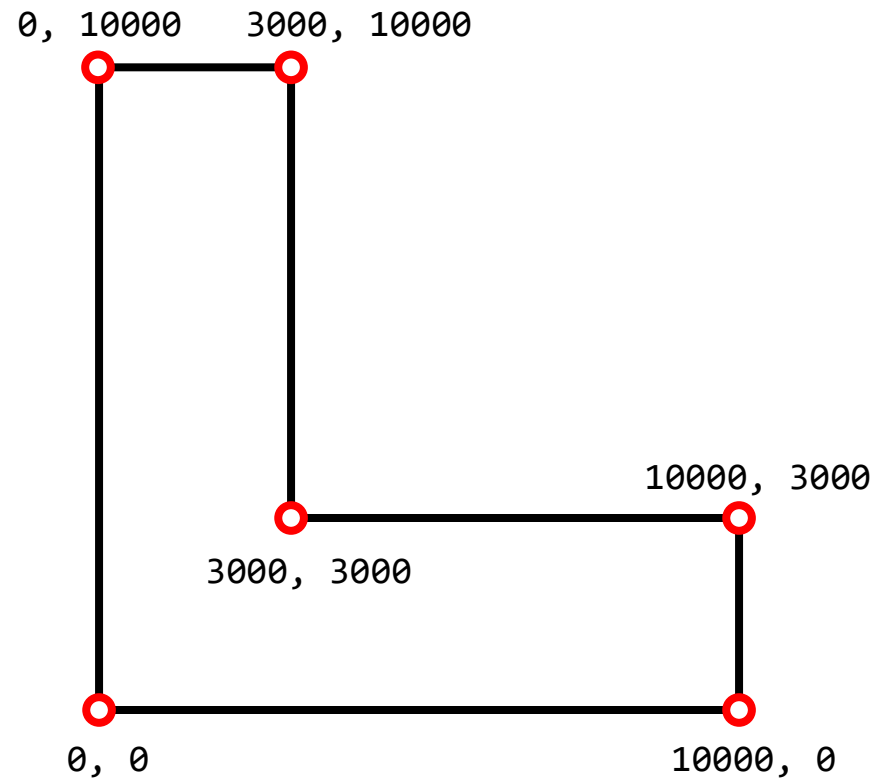


TYPES_OF_BAYS.CSV

INPUT: WAREHOUSE

Coord X, Coord Y

```
0, 0
10000, 0
10000, 3000
3000, 3000
3000, 10000
0, 10000
```

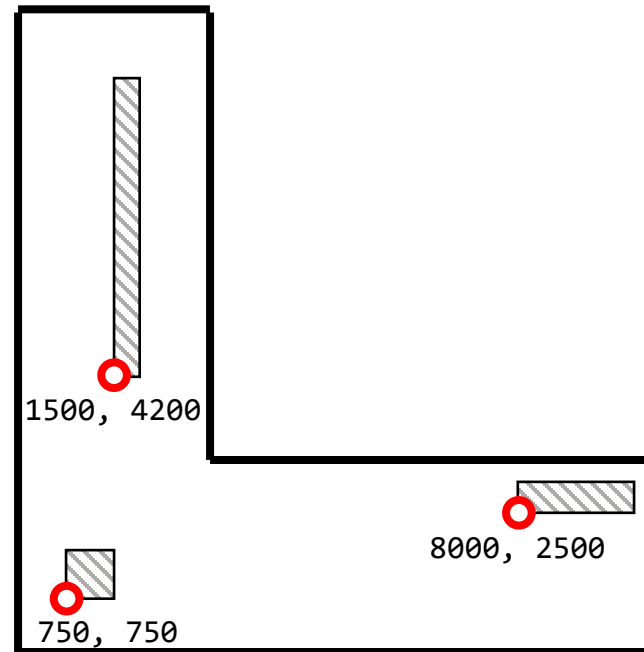


The walls of the warehouse
will always be axis aligned

INPUT: OBSTACLES

Coord X, Coord Y, Width, Depth

750, 750, 750, 750
8000, 2500, 1500, 300
1500, 4200, 200, 4600

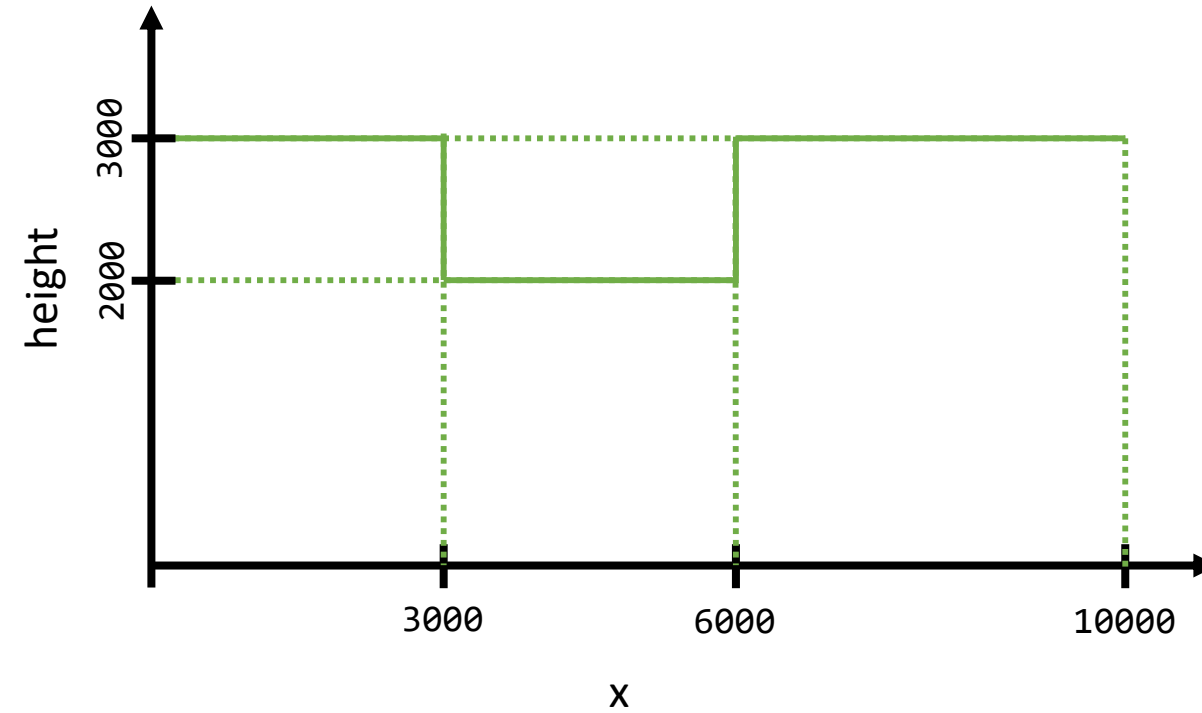


The obstacles will always be boxes

INPUT: CEILING

Coord X, Ceiling Height

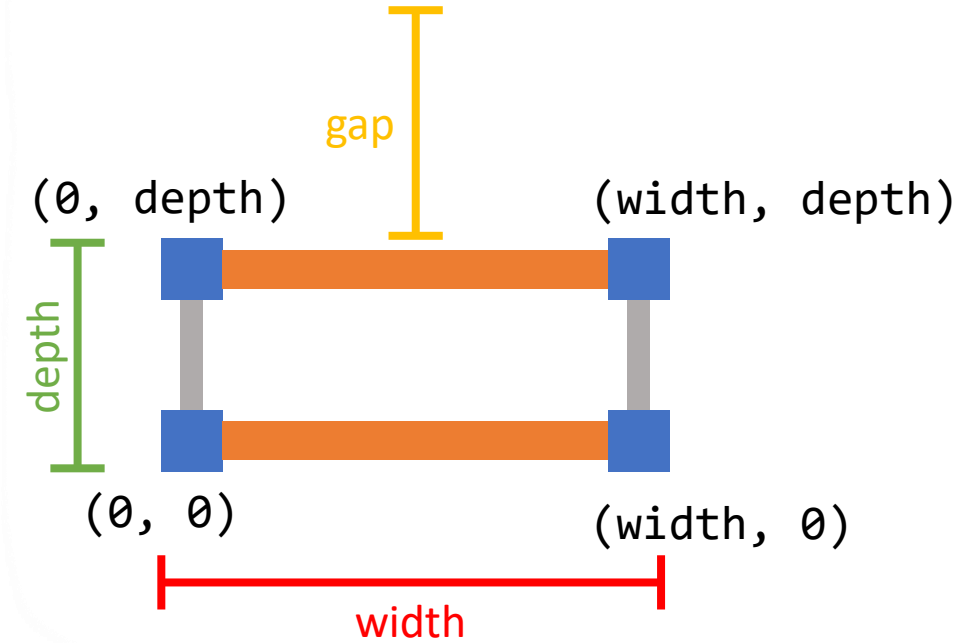
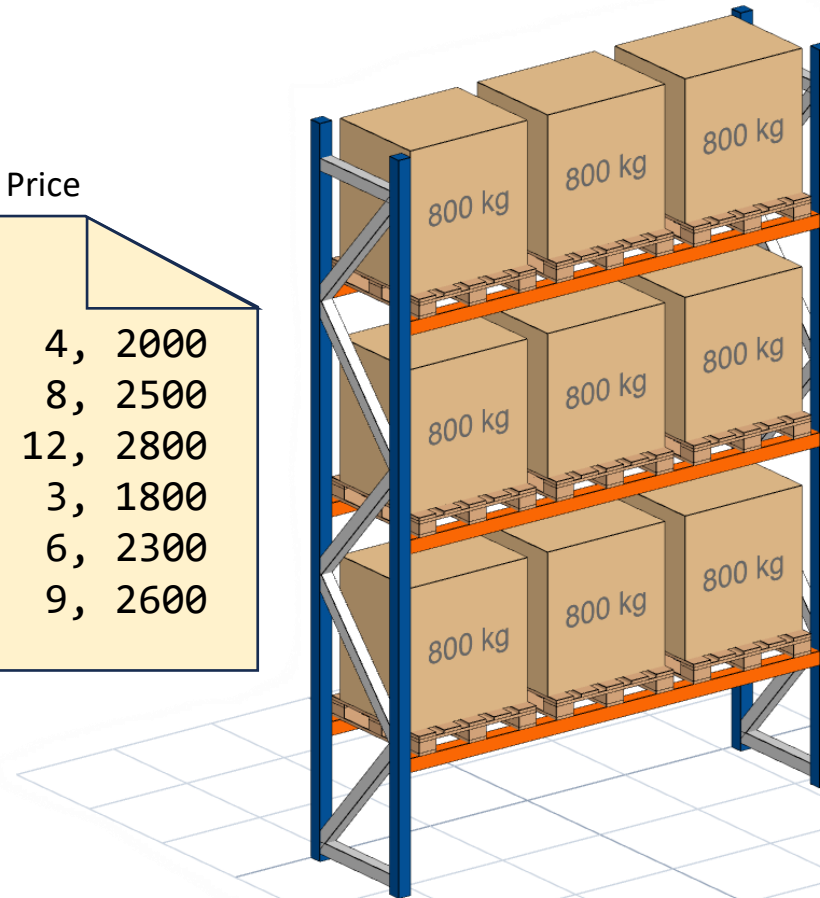
0, 3000
3000, 2000
6000, 3000



INPUT: BAY

Id, Width, Depth, Height, Gap, nLoads, Price

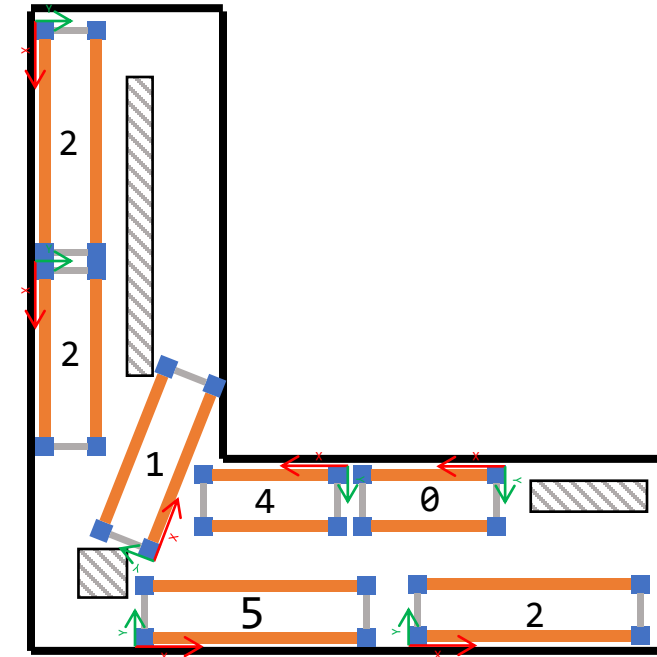
0,	800,	1200,	2800,	200,	4,	2000
1,	1600,	1200,	2800,	200,	8,	2500
2,	2400,	1200,	2800,	200,	12,	2800
3,	800,	1000,	1800,	150,	3,	1800
4,	1600,	1000,	1800,	150,	6,	2300
5,	2400,	1000,	1800,	150,	9,	2600



EXPECETED OUTPUT

Id, X, Y, Rotation

0	, 6600	, 2900	, 180
1	, 1000	, 1600	, 70
2	, 0	, 5400	, 270
2	, 0	, 9900	, 270
2	, 5000	, 0	, 0
4	, 4400	, 2900	, 180
5	, 2000	, 0	, 0



EVALUATION

Solution Generation

- Number of correct solutions
- Quality of the solution -> Minimize:

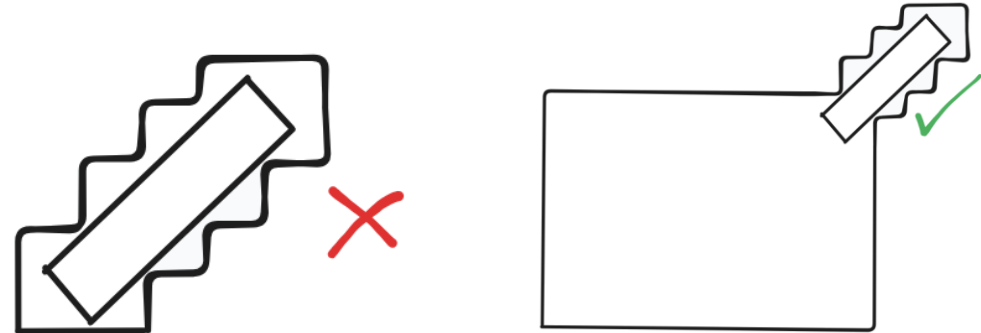
$$Q = \left(\sum_{bay} \frac{price}{loads} \right)^{2 - \frac{\sum_{bay} area}{area_{warehouse}}}$$

Presentation

- Visualization of the solution
- Whatever UI features

JUDGING

- During judging you will need to run some tests on your machine
 - Make sure it does not take more than 30s to finish
- Judging tests won't be crazy edge cases.
 - They may be bigger than the examples



- Bays can share boundaries, same with warehouse
 - If a bay at position (0,0) has width 1000, you don't need to put the next one at (1001,0), (1000,0) is fine.



ANY QUESTION?

GOOD HACKING